Berea Technical College
A LEADING FURTHER EDUCATION AND TRAINING PROVIDER.

MATHEMATICS N5

ASSIGNMENT 1

SUBJECT CODE: 16030175

Due Date: 31/08/2026

Examiner: Mr M Lazarus **Marks: 50**

Moderator: Mr Y Mia

INSTRUCTIONS AND INFORMATION

1. **Answer ALL the questions.**
2. **Read ALL the questions carefully.**
3. **All calculations must consist of the following:**
   - **3.1. A formula or the manipulation thereof.**
   - **3.2. Substitution of the given data in the formula**
   - **3.3. The answer together with the correct S.I. unit**
4. **Number all the answers correctly according to the numbering system used in the paper.**
5. **Formula sheet is provided.**
6. **Write neatly and legibly in blue or black ink only**

SPECIFIC INSTRUCTIONS AND INFORMATION

1. **All questions require full solution answers, complete the full solution for each question.**
2. **Do the required calculations on your A4 paper.**
3. **You need to submit your working out, you must submit ALL working out.**
4. **Formula sheet is attached.**
5. **Calculators may be used and you can refer to your study material.**
6. **A list of references must be provided for the assignment**
7. **Your assignment will include the attached cover page**

Question 1: Limits and Continuity [20]

1.1 Determine :$\lim_{x\to 0}\left(\frac{\ln(e^{2x}-3x)}{x}\right)$ (6)

1.2 $\lim_{x\to\infty}\left[\frac{4x^2+x-1}{3x^2+2x+1}\right]$.(2)

1.3 $\lim_{x\to 0}\left[\frac{1-e^x+x}{x^2}\right]$ (3)

1.4 $\lim_{x\to 0}\left[\frac{x-x\cos x}{x-\sin x}\right]$ (2)

1.5 Determine ln y if $\ln y=\lim_{x\to 3}\left[\frac{x^3-27}{x^2-9}\right]$ (2)

1.6 $\lim_{x\to 5}\left[\frac{x^3-125}{x-5}\right]$ (3)

1.7 Determine the values of x where $f(x)=\frac{x^2+x-6}{x^2+5x+6}$ is discontinuous. (2)

Question 2: Differentiation [30]

2.1 Differentiate f(x)=$\frac{2}{x}$ from first principles (4)

2.2 Determine $\frac{dy}{dx}$ if y=ln(2xlnx) (3)

2.3 Determine $\frac{dy}{dx}$ if y=2$\sin^{-1}\sqrt{x}$ +$\cot^{-1}\frac{x}{4}$ (3)

2.4 Calculate with the aid of logarithmic differentiation $\frac{dy}{dx}$ if $y=\left(\sqrt{x}\right)^{\cos x}$ (3)

2.5 Calculate by making use of the derivatives of $\sin x$ and $\cos x$, as well as the rules of differentiation: $y=\sec x$ (3)

2.6 Calculate $\frac{dy}{dx}$ if $f(x)=\sin x$ by means of first principles. Hint: $\lim_{h\to 0}\frac{\sin h}{h}$ (4)

2.7 Determine the slope of the tangent lines at the point (1: 2) of the graph:
$x^2+xy+y^2-5x=2$ (3)

2.8 Determine $\frac{dy}{dx}$ in each of the following cases (SIMPLICATION NOT REQUIRED)

2.8.1 $y=\sqrt{\ln x}-\log_7\tan x$ (3)

2.8.2 $y=arc\tan\left(\frac{1+x}{1-2x}\right)$ (2)

2.8.3 $y=x^3\ln(\cos 2x)$ (2)

TOTAL 50 MARKS